



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECR2523-01 DATABASE (WBL)

LAB ASSIGNMENT (DDL & DML)

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1.0 Data Definition Language (Loan Application Module)

1.1 CREATE TABLE status

```
CREATE TABLE `status` (  
  `status` int(3) NOT NULL PRIMARY KEY,  
  `statusDesc` varchar(25) NOT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Server: 127.0.0.1:3307 » Database: db_labdb » Table: status

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[Table structure](#) [Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	status	int(3)			No	None			Change Drop More
☐ 2	statusDesc	varchar(25)	utf8mb4_general_ci		No	None			Change Drop More

1.2 CREATE TABLE loanType

```
CREATE TABLE `loantype` (  
  `loanType` int(7) NOT NULL PRIMARY KEY,  
  `loanName` varchar(30) NOT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Server: 127.0.0.1:3307 » Database: db_labdb » Table: loantype

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#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	loanType	int(7)			No	None			Change Drop More
☐ 2	loanName	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More

1.3 CREATE TABLE loan

```
CREATE TABLE `loan` (
  `loanID` int(5) NOT NULL PRIMARY KEY,
  `loanAmount` decimal(10,2) NOT NULL,
  `loanDuration` int(2) NOT NULL,
  `loanStatus` int(3) NOT NULL,
  `monthlyPayment` decimal(10,2) NOT NULL,
  `bankName` varchar(30) NOT NULL,
  `accountBankNumber` bigint(25) NOT NULL,
  `applicantNetSalary` decimal(10,2) NOT NULL,
  `applicantBasicSalary` decimal(10,2) NOT NULL,
  `pengesahanMajikan` longblob NOT NULL,
  `loanApproveDate` date DEFAULT NULL,
  `staffNo` int(5) NOT NULL,
  `loanType` int(7) NOT NULL,
  `otherLoan` varchar(25) DEFAULT NULL,
  `loanReviewDate` date DEFAULT NULL,
  `loanApplyDate` date NOT NULL,
  `adminStaffNo` int(5) DEFAULT NULL,
  `alkStaffNo` int(5) DEFAULT NULL,
  FOREIGN KEY (loanStatus)
    REFERENCES status (status),
  FOREIGN KEY (staffNo)
    REFERENCES applicant (staffNo),
  FOREIGN KEY (loanType)
    REFERENCES loantype (loanType),
  FOREIGN KEY (adminstaffNo)
    REFERENCES admin (staffNo),
  FOREIGN KEY (alkstaffNo)
    REFERENCES alk (staffNo)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Server: 127.0.0.1:3307 » Database: db_labdb » Table: loan

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#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	loanID	int(5)			No	None			Change Drop More
☐ 2	loanAmount	decimal(10,2)			No	None			Change Drop More
☐ 3	loanDuration	int(2)			No	None			Change Drop More
☐ 4	loanStatus	int(3)			No	None			Change Drop More
☐ 5	monthlyPayment	decimal(10,2)			No	None			Change Drop More
☐ 6	bankName	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
☐ 7	accountBankNumber	bigint(25)			No	None			Change Drop More
☐ 8	applicantNetSalary	decimal(10,2)			No	None			Change Drop More
☐ 9	applicantBasicSalary	decimal(10,2)			No	None			Change Drop More
☐ 10	pengesahanMajikan	longblob			No	None			Change Drop More
☐ 11	loanApproveDate	date			Yes	NULL			Change Drop More
☐ 12	staffNo	int(5)			No	None			Change Drop More
☐ 13	loanType	int(7)			No	None			Change Drop More
☐ 14	otherLoan	varchar(25)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 15	loanReviewDate	date			Yes	NULL			Change Drop More
☐ 16	loanApplyDate	date			No	None			Change Drop More
☐ 17	adminStaffNo	int(5)			Yes	NULL			Change Drop More
☐ 18	alkStaffNo	int(5)			Yes	NULL			Change Drop More

1.4 CREATE TABLE guarantorinfo

```
CREATE TABLE `guarantorinfo` (  
  `guarantorIC` varchar(12) NOT NULL,  
  `guarantorName` varchar(100) NOT NULL,  
  `guarantorID` int(5) NOT NULL,  
  `guarantorPF` int(5) NOT NULL,  
  `guarantorPhoneNumber` varchar(11) NOT NULL,  
  `photo` longblob NOT NULL,  
  `loanID` int(5) NOT NULL,  
  PRIMARY KEY (guarantorIC, loanID)  
  FOREIGN KEY (loanID)  
    REFERENCES loan (loanID)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Server: 127.0.0.1:3307 » Database: db_labdb » Table: guarantorinfo

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[Table structure](#) [Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 guarantorIC 🔑	varchar(12)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 guarantorName	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 guarantorID	int(5)			No	None			Change Drop More
☐	4 guarantorPF	int(5)			No	None			Change Drop More
☐	5 guarantorPhoneNumber	varchar(11)	utf8mb4_general_ci		No	None			Change Drop More
☐	6 photo	longblob			No	None			Change Drop More
☐	7 loanID 🔑🔑	int(5)			No	None			Change Drop More

2.0 Data Manipulation Language (Loan Application Module)

2.1 SELECT membershipStatus, modahSyer

This sqlMembership function is to select membershipStatus from membership table and modahSyer from savingtype table. The membershipStatus of applicant should equal to 2 which is 'approved' and their modahSyer should be more or equal than 300 to apply for a loan.

```
$sqlMembership = "SELECT membership.membershipStatus, savingtype.modahSyer  
FROM membership  
JOIN applicant ON membership.staffNo = applicant.staffNo  
JOIN savingType ON savingType.staffNo = membership.staffNo  
WHERE applicant.email = '$ userEmail';"
```

Example of \$userEmail = nawwarahauni@graduate.utm.my. Hence, the output should be:

membershipStatus	modahSyer
2	300

2.2 SELECT applicant information and their respective bank account name and account bank number

This sql function is to select some of applicant information and bankAccountName as well as accountBankNumber from bankinfo table. All information selected is used to auto fill in for applicants when they fill in the 'Maklumat Pemohon' in loan application form so that they do not have to fill in their details again.

Example of \$userEmail = nawwarahauni@graduate.utm.my. Hence, the output should be:

```
$sql = "SELECT applicant.applicantName,
            applicant.applicantIC,
            applicant.applicantDOB,
            applicant.applicantAge,
            applicant.applicantStreet,
            applicant.applicantPostcode,
            applicant.applicantGender,
            applicant.applicantReligion,
            applicant.applicantRace,
            applicant.applicantPF,
            applicant.applicantPosition,
            applicant.officeStreet,
            applicant.officePostcode,
            applicant.applicantPhoneNumber,
            bankinfo.bankAccountName,
            bankinfo.accountBankNumber
        FROM applicant
        LEFT JOIN users ON users.email = applicant.email
        LEFT JOIN bankinfo ON bankinfo.staffNo = applicant.staffNo
        LEFT JOIN membership ON membership.staffNo = applicant.staffNo
        WHERE users.email = '$ userEmail';"
```

applicantName	applicantIC	applicantDOB	applicantAge	applicantStreet	applicantPostcode	applicantGender	applicantReligion	applicantRace	applicantPF	applicantPosition
NAWWARAH AUNI BINTI NAZRUDIN	40518160909	2004-05-18	21	JALAN KEMUBU 8/9	40320	P	ISLAM	MELAYU	5468	D

officeStreet	officePostcode	applicantPhoneNumber	bankAccountName	accountBankNumber
JALAN KEMUBU 8/9	40320	195143276	BANK RAKYAT	6472195734

2.3 INSERT INTO loan table

This function is to store the value inserted from the loan application form into the loan table. The otherLoan will not be NULL only if the applicant chooses loanType = 7 which is 'other'. Meanwhile, adminStaffNo and loanReviewDate will be filled later when the admin reviews the application and alkStaffNo and loanApproveDate will be filled later when the Ahli Lembaga Koperasi approves the application.

```
$sqlInsertLoan = "INSERT INTO loan
    (loanAmount, loanDuration, loanStatus, monthlyPayment, bankName, accountBankNumber, loanType, otherLoan,
    staffNo, adminStaffNo, loanApplyDate, loanReviewDate, loanApproveDate, applicantNetSalary,
    applicantBasicSalary, pengesahanMajikan)
VALUES ('$amount', '$duration', '1', '$mpayment', '$bank', '$accNum', '$type', '$other', '$staffNo', $
    adminStaffNo, CURRENT_TIMESTAMP(), $loanReviewDate, $loanApproveDate, '$netSalary', '$basicSalary', '
    $fileContent3')";
```

Example values:

\$amount = RM 10000,

\$loanDuration = 24 (in months),

\$loanStatus = 1,

\$mpayment = RM 434.17,

\$bank = RHB Bank,
 \$accnum = 76408751157,
 \$type = al-bai,
 \$other = NULL,
 \$staffNo = 1234,
 \$adminStaffNo = NULL,
 \$loanReviewDate = NULL,
 \$loanApproveDate = NULL,
 \$netSalary = 7511,
 \$basicSalary = 7890,
 \$fileContent3 = uploaded pdf.

Hence, the output should be:

loanID	loanAmount	loanDuration	loanStatus	monthlyPayment	bankName	accountBankNumber	applicantNetSalary
20	10000.00	24	1	434.17	RHB Bank	76408751157	7511.00

applicantBasicSalary	pengesahanMajikan	loanApproveDate	staffNo	loanType	otherLoan	loanReviewDate	loanApplyDate	adminStaffNo	alkStaffNo
7890.00	[BLOB - 13 B]	NULL	1234	1	NULL	NULL	2025-01-21	NULL	NULL

Function below is to store first guarantor information.

```

$sqlInsertGuarantor1 = "INSERT INTO guarantorInfo
                        (guarantorIC, loanID, guarantorName, guarantorID, guarantorPF, guarantorPhoneNumber, photo)
                        VALUES ('$gIC1', '$loanID', '$gName1', '$gID1', '$gPF1', '$gPhoneNumber1', '$fileContent1')";
  
```

Example values:

\$gIC1 = 710125101236,
 \$loanID = 20,
 \$guarantorName = Qawie binti Awie,
 \$guarantorID = 1236,
 \$guarantorPF = 6321,
 \$guarantorPhoneNumber = 01666632222,
 \$fileContent1 = uploaded pdf.

Function below is to store second guarantor information.

```

$sqlInsertGuarantor2 = "INSERT INTO guarantorInfo
                        (guarantorIC, loanID, guarantorName, guarantorID, guarantorPF, guarantorPhoneNumber, photo)
                        VALUES ('$gIC2', '$loanID', '$gName2', '$gID2', '$gPF2', '$gPhoneNumber2', '$fileContent2')
                        ";
  
```

Example values:

\$gIC1 = 7512060123457,
 \$loanID = 20,

\$guarantorName = Muhammad bin Ahmad,
 \$guarantorID = 3457,
 \$guarantorPF = 7543,
 \$guarantorPhoneNumber = 01777754444,
 \$fileContent1 = uploaded pdf.

Hence, the output should be:

guarantorIC	guarantorName	guarantorID	guarantorPF	guarantorPhoneNumber	photo	loanID
710125101236	Qawie binti Awie	1236	6321	01666632222	[BLOB - 13 B]	20
751206012345	Muhammad bin Ahmad	3457	7543	01777754444	[BLOB - 13 B]	20

2.4 SELECT loanID, loanStatus, loanApplyDate, loanApproveDate, loan.loanType, loantype.loanType, loanType.loanName.

This function should retrieve some of the loan information to be used to update the status of the loan application.

```
$query = "SELECT loanID, loanStatus, loanApplyDate, loanApproveDate, loan.loanType, loantype.loanType, loanType.loanName
FROM loan
LEFT JOIN applicant ON loan.staffNo = applicant.staffNo
LEFT JOIN loantype ON loantype.loanType = loan.loanType
WHERE applicant.email = '$ userEmail'
ORDER BY loanApplyDate DESC";
```

Example of \$userEmail = nawwarahauni@graduate.utm.my. Hence, the output should be:

loanID	loanStatus	loanApplyDate	loanApproveDate	loanType	loanType	loanName
20	1	2025-01-21	NULL	1	1	al-bai
5	2	2025-01-09	2025-01-21	6	6	cukai jalan

2.5 UPDATE applicant

This function is to allow users to update their profile if they want to make some changes.


```
$updateSql = "UPDATE applicant SET
    applicantReligion = '$applicantReligion',
    applicantRace = '$applicantRace',
    applicantPosition = '$applicantPosition',
    applicantDOB = '$applicantDOB',
    applicantAge = '$applicantAge',
    applicantStreet = '$applicantStreet',
    applicantPostcode = '$applicantPostcode',
    applicantCity = '$applicantCity',
    applicantState = '$applicantState',
    applicantPhoneNumber = '$applicantPhoneNumber'
    WHERE email = '$userEmail'";
```

Example values:

\$applicantReligion = Islam,
 \$applicantRace = Cina,
 \$applicantPosition = Ahli Lembaga Koperasi,
 \$applicantAge = 40,
 \$applicantStreet = Jalan Lorong Berkat 1,
 \$applicantPostcode = 47100,
 \$applicantCity = Puchong,
 \$applicantState = Selangor,
 \$applicantPhoneNumber = 01117371502,
 \$userEmail = nawwarahauni@graduate.utm.my.

Hence, the output should be:

applicantReligion	applicantRace	applicantPosition	applicantAge	applicantStreet	applicantPostcode	applicantCity	applicantState	applicantPhoneNumber
Islam	Cina	Ahli Lembaga Koperasi	40	Jalan Lorong Berkat 1	47100	Puchong	Selangor	01117371501